A proposed 389-variable quartic refinement

Reusing primary depth, refining the height estimate,
and reducing coefficients by rational scaling

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1 Selected result and its mathematical scope

The selected polynomial has **389 total rational variables**, including an unchanged distinguished input X, and therefore 388 existential witnesses. It is the sum of squares of 357 explicit integer quadratic residuals. Its degree is exactly four, it has 26,295 nonzero monomials, all 389 variables occur, and the coefficient of X^4 is one. In particular its integer content is one. Every coefficient in the supplied text table is an ordinary decimal integer. The largest absolute coefficient is

$$\boxed{28902 \cdot 3^{1576}}, \qquad \textbf{757 decimal digits and 2513 bits}.$$

This replaces the preceding 392-variable proposal, which had 360 residuals, 26,301 monomials and a 905-digit maximum coefficient. No extra variable is used to name the fixed constant in the selected form.

The intended characterization, with the inherited global arithmetic qualifications retained, is

$$X \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists (y_1, \ldots, y_{388}) \in \mathbb{Q}^{388} \quad P_{389}(X, y) = 0.$$

The finite algebraic construction and the new local arguments are supplied and checked here. This proposal does not independently establish all of the inherited curve/group, quaternionic, modularity and local compatibility, reduction, isogeny-height, simultaneous prime/progression, Lucas and pole realization inputs. In particular the improved height exponent below is a *reapplication of the inherited proof inputs*, with sharper explicit estimates; it is not inferred merely by replacing the exponent in an existing theorem statement. Internal reviews by separately assigned AI agents do not constitute human peer review or a completed Lean proof.

1.1 Three selected improvements

The detector change promotes the primary inverse-root test $g(U_A)$ from order one to order twelve. Completeness already puts H at formal depth at least 45, and chooses $U_A = [n]H$ with $n \in \mathbb{Z}$. Integer multiplication cannot lower that positive formal depth. The new uniform pair uses the already stored square q_{12}^2, deleting the three separate products $q_1 q_{12}$, one per detector. Soundness uses only the strengthened local conclusion and does not assume the completeness relation for an arbitrary putative solution.

The arithmetic change sharpens an Euler-product estimate and uses $g \geq 2$ with $g \leq 1 + d_H/12$ to obtain $g \leq d_H/6$. These give the new height exponent $\alpha = 577/32 = 18.03125$ and the 665-bit constant

$$\mathcal{H} = 3 \cdot 2^{106} (2^{32} 3^9)^6 46410^{18} + 128.$$

Checking the actual scalar, Lucas and input consumers permits the smaller fixed anchor $S_0 = 3^{2364}$, in place of 3^{2790}. In particular the scalar target is $h(\beta) < |a-1|/4$, which the final input contradiction actually needs. Scalar order 28, high order 45, input order 12, packing coefficient three and the existing prime-selection hypotheses remain.

The coefficient change combines the preceding ternary scaling with two additional dyadic factors. It changes thirteen existing coordinates by fixed nonzero rational factors and fixes X. Exact rational row normalization then gives the stated 757-digit maximum. The negative dyadic exponent creates five nonintegral row divisors, all explicitly certified. The square-scale hypotheses apply to the decoded old semantic scale; the new stored scale coordinate need not itself be a square.

1.2 Available coefficient-size tradeoffs

The construction also regenerates the direct unbalanced polynomial and three constant-chain variants. All counts below include the input. Each constant-chain variant is a unique extension of the same source system; the added coordinates are counted.

Form	Variables	Residuals	Monomials	Max. digits
Selected balanced decimal	**389**	357	26,295	**757**
Direct unbalanced decimal	389	357	26,295	2,273
Two-coordinate constant lift	391	359	26,304	752
Three-coordinate constant lift	392	360	26,307	323
Four-coordinate constant lift	393	361	26,310	151

Thus the new three-coordinate lift has the preceding proposal's total variable count 392 with a 323-digit coefficient maximum. The four-coordinate lift uses fewer variables than the preceding two-coordinate lift and has much smaller coefficients. These are measured constructions, with no claim that any point of this tradeoff is optimal.

1.3 Provenance and how to read this note

The upstream repository is https://github.com/elliotglazer/diophantine-z-q, pinned at `036ce be4f1a4aa7ca89820a9b48f3629b4c364d6`. The pin is a reproducibility reference, not a claim about current upstream HEAD. The proposal is an additive draft under `research/conditional-quartic-reduction/`; it is not an official release or a designation of the next manuscript version.

The following three sections are the complete new argument texts. The detector section first measures its isolated change at the preceding anchor; the arithmetic section separately justifies the selected smaller anchor; the diagonal section applies to the integrated result. The exact integrated source, proof texts and receipts are bound in `PROVENANCE.json` and `SOURCE_MANIFEST.json`.

The unchanged seventh-round detector normalization and scalar proof, and the earlier local/arithmetic derivations on which they depend, remain in the supplied proof tree. The preceding compiled note is retained under `lineage/round7_proposal_note.pdf`. The full dependency inventory identifies the actual domain and consumer of every retained transformation; historical numerical examples are not selected current assumptions.

2 Use available primary depth to share the order-twelve square

The frozen preceding source has SHA256 `78b54bcde24ff8a91091e102a4734c49878699f071d9230e5ce7a70ffd2114d0`. The isolated replacement in this section has SHA256 `5eae7a6cb6fd22e2da8a4573 54628355ece7e8930b60200d088cd0cfca1d28cb`. It saves one rational coordinate in each of the three detectors by strengthening one test which the existing completeness witnesses already satisfy to higher order. No global arithmetic input is proved here; their qualifications and the preceding complete local arguments are retained.

2.1 The actual test whose order changes

Write $U_A = [u_A]P_*$ for the primary point attached to the input label, $g_A = g(U_A)$ for its complete reciprocal-chart coordinate, and

$$D_\times = L_1 - \overline{\ell(V_1)/\ell(V_0)}$$

for the existing input cross test. Thus $g_A^2 = f(U_A)$, and the source aliases are `test1` and `test10`, respectively. The preceding detector required orders one and twelve on this pair. Require order twelve on *both* tests instead. The complete order list, in the source order `test0`,...,`test15`, is now

$$(45, 12, 1, 1, 1, 1, 1, 1, 1, 1, 12, 45, 45, 45, 45, 45).$$

There are eight order-one tests, two order-twelve tests, and six order-forty-five tests. All underlying field functions, their conjugations, and all other orders are unchanged. In particular no order-one alternate point test is strengthened: the alternate points still need only formal depth one.

The retained one-place lemma applies to arbitrary positive orders. With $t = A\lambda$, $\delta = 98t - 1$, and

$$R = \frac{F(t)}{A^5\delta^6}, \qquad q_m = \frac{e^{2m-1}}{\delta}, \qquad e = \frac{(A\delta)^5}{F(t)},$$

the semantic test is still δD, and the implemented factor is $1 + q_m D^2$. At a consumed contact δ is a unit, so the common-place conclusion for the new order list supplies $v_{\mathfrak{p}}(g_A) \geq 12$. This implies the former order-one conclusion and leaves every other required conclusion unchanged. The subsequent primary/alternate comparison, the actual G_2/G_{12} error bounds, the identity-input case, and the three height consumers therefore apply with their old premises. This is a strengthened local predicate used in the conditional soundness argument; it is not a bijection between every pair of old and new detector witness tuples.

2.2 Why completeness already gives this stronger primary test

In the retained completeness construction, the integer input X first fixes the integer label

$$n = \beta(X) = 11016X^2 + 11016X + 5155.$$

The primary points are chosen as $H = [h]P_*$ and $U_A = [hn]P_* = [n]H$. At every selected contact place used for the primary side, the existing progression $\Pi^{44} \mid h$, together with the prescribed annihilating reduction order, makes H formal of depth at least forty-five. This is exactly the existing depth used for the high-order `test0`; the present change imposes no new congruence on h.

For completeness of this deduction, let E_s be the subgroup of a good integral formal group whose parameter z has valuation at least the positive integer s. The integral formal addition and inverse series show that E_s is a group. Hence $[n]E_s \subseteq E_s$ for every integer n, including negative n and

$n = 0$. Equivalently the integral series $z([n]Q) = nz(Q) + O(z(Q)^2)$ cannot lower its positive valuation. Therefore

$$v_{\mathfrak{p}}(z(U_A)) \geq v_{\mathfrak{p}}(z(H)) \geq 45.$$

On a formal point in the reciprocal chart, $z = -g/l$ and $l^2 = 1 - Dg^4$; the relevant l is a unit. Consequently

$$v_{\mathfrak{p}}(g_A) = v_{\mathfrak{p}}(z(U_A)) \geq 45 \geq 12.$$

Both choices of the square-root chart have the same valuation. The identity case has $g_A = 0$ and satisfies every requested positive order; no nonidentity guard for U_A has been inserted. The consumed primes remain outside $\{2, 3, 5, 7, 13, 17\}$, so the good integral and unramified formal-group hypotheses are exactly those already used in the retained primary-depth argument.

The existing CRT conditions and their coprimalities are unchanged. In particular the alternate condition involving $h - 1$ is not strengthened, and one does not demand that a selected prime divide both h and $h - 1$. The scalar preparation and continuity neighborhoods, root points, square scales and finite valuation/real bounds are still fixed before the asymptotic contact-prime selection. This argument uses the actual completeness choice $U_A = [n]H$; it does not assume that equality for an arbitrary putative solution in the soundness direction.

2.3 The shared uniform block and its exact cost

Put $U = g_A^2$ and $V = D_\times^2$, both elements of $K = \mathbb{Q}(\sqrt{2})$. The preceding product block was

$$(1 + q_1 U)(1 + q_{12} V) = 1 + q_1 U + q_{12} V + (q_1 q_{12}) UV.$$

The replacement is the uniform-order block

$$(1 + q_{12} U)(1 + q_{12} V) = 1 + q_{12}(U + V) + q_{12}^2 UV.$$

The shared field coefficient UV is exactly the same as before. The stored square q_{12}^2 is already the coordinate `detector*.mixedHigh`, required by the unchanged high-weight relation

$$q_1^3 q_{45} = (q_{12}^2)^2.$$

Thus each detector can delete its separate coordinate for $q_1 q_{12}$. The two rational output coordinates of the block remain; every defining row is quadratic in the stored coordinates. The two four-test order-one blocks and two three-test high-order blocks use exactly their preceding lists, because the promoted test was precisely the one previously removed from the order-one list for the mixed pair. Their shared coefficients, product evaluations and weight powers are unchanged. The seven-node higher-weight chain also remains unchanged.

On the positive source domain $F, q_1, q_{12}, q_{45}, E$ are positive. Both squared tests are nonnegative at either real embedding. The new block and each other factor are therefore at least one. The old conclusions $W_0 \geq 1$, $N(W) > 0$, nonzero reciprocal trace, and positive pole input remain valid. No quotient denominator or branch restriction has been added. In the construction of the weight chain, all denominators remain the previously checked nonzero E and q_1^3.

2.4 The changed factor in the pole-completeness estimates

At a selected contact prime, $v_p(R) = 1$ and δ is a unit. At the successful place, $v(g_A) \geq 45$ gives

$$v(q_{12} g_A^2) = 1 - 24 + 2v(g_A) \geq 67 > 0.$$

The new factor is thus one modulo the prime. At either place, a nonzero nonconstant term has odd valuation. Each inverse factor has residue zero or one, and the reciprocal trace has the same unit residue one or two as in the preceding proof. At noncontact primes the even valuation of the unchanged R supplies the same square-denominator argument for the pole input.

At real infinity every fixed-order weight still satisfies $q_m = O(\lambda^{-1})$. The underlying g_A is one of the already uniformly bounded tests in the real progression argument, so replacing $q_1 g_A^2$ by $q_{12} g_A^2$ changes only its fixed bound. The full product still satisfies $W = 1 + O(\lambda^{-1})$ and the same pole input tends to zero through positive values. In particular the required bound $B < 17$ is eventually satisfied.

At $p = 2, 3, 17$, retain $\nu = v_p(v)$, $\eta = v_p(A)$ and $\alpha = v_p(98)$. The preceding exact weight estimate is

$$v_p(q_m) = \nu + (10m - 6)(\eta + \alpha).$$

For the promoted test and its fixed lower bound $v(g_A) \geq -c_p$ this becomes

$$v_p(q_{12} g_A^2) \geq \nu + 114(\eta + \alpha) - 2c_p \longrightarrow +\infty.$$

Thus the same freedom to increase the denominator divisibility supplies the preparation-zero pole conditions. The scale, bound c_p, and order twelve are fixed before this choice. The argument does not require $\eta + \alpha$ to be nonnegative. Together with the unchanged local squareclasses and pole-input formula, this verifies the new completeness application.

2.5 Measured construction and verification scope

With all other seventh-round parameters retained, including $S_0 = 3^{2790}$, full compilation gives

$$\text{400 coordinates/368 residuals} \quad \longrightarrow \quad \boxed{\text{389 coordinates/357 residuals}}.$$

The source has 339 gates. The independent gate checker verifies all 339 identities and 330 compiler operations. The compiled polynomial has degree four, 26,295 nonzero monomials, primitive content one, and all 389 coordinates active. These counts include the distinguished input, which remains unchanged. The three-coordinate saving and six-monomial decrease are relative to the frozen 392-coordinate, 360-residual construction; no final-elimination saving is counted twice.

The separately executed source checker `check_detector_round8.py` imports neither producer nor compiler. It reconstructs all sixteen semantic tests and the new order allocation, the unchanged logarithms, weight chain, shared coefficients, uniform pair, remaining products, norm/pole consumers and lifted degree. It passes 384 source assertions, seven Laurent identities, three normalization identities and two expanded component identities for the uniform pair. These exact finite checks bind the implementation; the formal-group and completeness arguments above supply the uniform mathematical deduction. All inherited global arithmetic remains conditional as in the preceding package.

3 A smaller height exponent and an anchor of exponent 2364

This argument starts from the frozen seventh-round source with builder SHA256 `78b54bcde24ff8a910 91e102a4734c49878699f071d9230e5ce7a70ffd2114d0`. The upstream manuscript remains pinned at `036cebe4f1a4aa7ca89820a9b48f3629b4c364d6`. The new assertions below are explicit applications of the same inherited geometric, modularity, isogeny-height, group, quaternion, pole, prime-pattern and Lucas inputs. They do not independently establish those inputs or reclassify any historical omission as a new gap.

The selected arithmetic change in the source is only $S_0 = 3^{2790} \rightsquigarrow 3^{2364}$. The physical anchor's packing coefficient stays equal to three. The local conclusions consumed here remain scalar order 28, Lucas-label order 45 and input-label order 12. A strengthened primary detector test, if selected elsewhere in this proposal, has its own local/completeness proof and does not weaken any of these conclusions. The current integrated source and this proof are bound by the accompanying independent checker report.

3.1 The precise inherited height inputs

Put

$$\mathcal{S} = \{2, 3, 5, 7, 13, 17\}, \qquad B_{\mathcal{S}} = 46410, \qquad C_{\mathrm{cond}} = 2^{32}3^9.$$

For $z \in \mathbb{Q} \setminus \{0, 1, 2401, -343/32\}$, let $M_{\mathcal{S}}(z)$ be the squarefree product of primes outside $\mathcal{S}$ having odd contact with one of $\infty, 0, 1, 2401, -343/32$, and set $R = B_{\mathcal{S}} M_{\mathcal{S}}(z)$. Thus $R \geq 1$ is a squarefree rational integer. This R is the radical used in the global height proof; it is not a detector pole input or an elliptic-curve index.

The finite-support reapplication in `proofs/prior/round4/arithmetic_round4.tex` and the conductor refinement in `proofs/prior/round5/arithmetic_round5.tex` retain the source's geometric construction and (LR1)–(LR5). In particular, in the non-CM case they supply the same surface A, field L', primitive newform of level N, and finite nebentypus of order t, with

$$d' = [L' : \mathbb{Q}] \leq 2304R, \qquad N \leq C_{\mathrm{cond}} R^2, \qquad \mathrm{rad}(N) \mid R, \qquad t \leq 2d' \leq 4608R.$$

The last estimate is a character-order bound, not a coefficient-field degree bound. The proofs of normality, the scalar-character corrections, the companion at three used to bound the conductor at five, and the conductor's primitive-level convention remain inherited and unchanged.

The actual source is `manuscripts/quartic_v10.tex`, `sec:h5-level-repair` and `thm:quartic-heigh t19`, especially lines 8915–9073. Its subgroup $H = \ker \epsilon$ contains $-I$ and gives a modular curve X_H over $\mathbb{Q}$, with

$$d_H = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_H] = Nt \prod_{p \mid N} (1 + 1/p), \qquad \deg_B(X_H) \leq d_H, \qquad g \leq 1 + d_H/12.$$

Here $g = \dim \mathrm{Jac}(X_H)$. The inherited realization, absolute simplicity and Faltings–Poincare argument put the two-dimensional surface A in that Jacobian up to isogeny over L'; therefore $g \geq 2$. No bounded coefficient-field degree, squarefree level, or trivial nebentypus is being introduced.

3.2 An elementary Euler-product refinement

Let

$$\mathcal{P} = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}.$$

These are exactly the primes below 53. For $a_p = (1+1/p)p^{-1/192}$, the following finite integer inequalities hold:

$$(p+1)^{192} > p^{193} \qquad (p \in \mathcal{P}),$$
$$54^{192} < 53^{193},$$
$$\left(9\prod_{p\in\mathcal{P}}(p+1)\right)^{192} < 32^{192}\left(\prod_{p\in\mathcal{P}} p\right)^{193}.$$

The companion certificate verifies every integer comparison, as well as the completeness and primality of the displayed list. The function $(1+1/x)^{192}/x$ is decreasing for $x > 0$, as a product of positive decreasing functions. Hence $a_p \le 1$ for every prime $p \ge 53$, whereas every displayed small-prime factor exceeds one. For every positive integer N it follows that

$$\prod_{p\mid N}(1+1/p) = \operatorname{rad}(N)^{1/192}\prod_{p\mid N} a_p \le \operatorname{rad}(N)^{1/192}\prod_{p\in\mathcal{P}} a_p < \frac{32}{9}\operatorname{rad}(N)^{1/192}.$$

This controls arbitrary subsets of the small primes; a product of unconditionally smaller-than-one factors has not been inserted.

Define

$$\alpha = 18 + \frac{1}{32} = \frac{577}{32}, \qquad T = 2^{14}C_{\mathrm{cond}}R^{577/192}.$$

The level, character and radical bounds above now give

$$d_H < 4608\,\frac{32}{9}C_{\mathrm{cond}}R^{3+1/192} = T.$$

The number 53 is only a cutoff in a numerical Euler-product estimate. It does not enlarge $\mathcal{S}$, discard contacts at smaller primes, or alter any local logarithm, pole, ramification or prime-selection hypothesis.

3.3 Reapply the Jacobian and isogeny-height bounds

The genus estimate can be used without the source's coarse replacement $g \le d_H$. Indeed $g \ge 2$ and $g \le 1 + d_H/12$ imply $d_H \ge 12$, and hence

$$g \le 1 + d_H/12 \le d_H/6 < T/6.$$

Use the same stable Faltings metric as the source. Its cited Javanpeykar bound is

$$h_F(J_H) \le 13000000\,\deg_B(X_H)^5 g < \frac{13000000}{6}T^6.$$

The source's Gaudron–Rémond application to the principally polarized J_H and its isogeny factor $A \times Z$ gives

$$h_F(A) + h_F(Z) \le h_F(J_H) + V,$$
$$V = 2g^2\left(8g^2\log(7g) + \log d' + \log\max(1, \log d', h_F(J_H))\right).$$

Retain its already proved upper estimate, with $u = \log T$:

$$V \le 16T^4(2+u) + 2T^2(25+7u) \le T^6.$$

All hypotheses used for this estimate still hold: $T \ge 16$, $g < T$, $\log d' < 8 + \log R \le 8 + u$, and the new Jacobian bound is smaller than the source's $2^{24}T^6$ bound. The elementary inequalities $2 + \log T \le T^2/32$

and $25 + 7\log T \le T^4/4$ hold at 16 and persist by differentiation, exactly as in the source. No new height-variation theorem is used.

The retained Bost lower bound gives $h_F(Z) > -3g/2 > -3T/2$. Since $3T/2 < T^6$ for $T \ge 16$ and

$$\frac{13000000}{6} + 2 < 2^{22},$$

we obtain

$$h_F(A) < 2^{22}T^6 = 2^{106}C_{\mathrm{cond}}^6 R^\alpha.$$

The original CM estimate $h_F(A) < 854R$ is smaller than this bound as well. Applying the same geometric comparison $h(z) \le 2\max(0, h_F(A)) + 128$ gives

$$h(z) < 2^{107}C_{\mathrm{cond}}^6 R^\alpha + 128.$$

Finally the exact comparison $2^{32}B_{\mathcal{S}} < 3^{32}$ implies $B_{\mathcal{S}}^{1/32} < 3/2$. Thus, with the integer

$$\mathcal{H} = 3 \cdot 2^{106}C_{\mathrm{cond}}^6 B_{\mathcal{S}}^{18} + 128, \tag{1}$$

and using $M_{\mathcal{S}}(z) \ge 1$, the reapplication proves

$$\boxed{h(z) \le \mathcal{H}\, M_{\mathcal{S}}(z)^{577/32}.} \tag{2}$$

The harmless weak inequality is convenient for all subsequent consumers. The exact integer $\mathcal{H}$ has bit length 665 and satisfies $\mathcal{H} < 3 \cdot 2^{663}$. If H_7 denotes the preceding 683-bit constant, then

$$H_7 - 128 = 247520(\mathcal{H} - 128).$$

This smaller exponent and coefficient follow from the displayed inequalities in the inherited proof, rather than from a formal substitution into its final exponent-nineteen theorem statement.

3.4 The retained even anchor and its noncircular local order

Write $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$ and $N_K(x + y\rho) = x^2 - 2y^2$. Let the four root labels be (n, a, t, b), and put

$$C = \prod_i x(Q_i) = C_0 + C_1\rho, \qquad B_0 = 4096C_0.$$

The scalar point is $S = [a]P$, its shifted point is $Q = S - P = [a-1]P$, and the retained chart and nonzero guard give $a \ne 0$, $a - 1 \ne 0$, and an existing field coordinate $w = h_S\tau$ satisfying $h_S^2 = x(S)$ and $\tau^2 = f(Q)$. Set

$$S_0 = 3^{2364}, \qquad \Theta = N_K(w) + 3S_0N_K(B_0 + X\rho),$$

and retain the three-witness conic

$$u^2 + 6v^2 - 2z^2 - 3\Theta^2 + 2S_0^2 = 0.$$

Its exact point/input domain is

$$\text{all four labels odd}, \qquad X \in \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}, \qquad 0 \ne a - 1 \in 2 \cdot 3^{1181}\mathbb{Z}. \tag{3}$$

Here is the local order, making explicit why the parity argument is not circular. The same quaternion criterion, including its zero argument, tests 2- and 3-integrality of

$$\Theta/S_0 = N_K(w/3^{1182}) + 3N_K(B_0 + X\rho).$$

At 3 the norm is anisotropic modulo 3, so the two nonzero summands have opposite valuation parity and must separately be integral. This gives integrality of B_0, X and depth at least 1182 for both coefficients of w. Before assuming oddness, the retained leading-residue proof gives

$$v_3(C_0) = -\sum_i e(n_i), \qquad e(n) = 0 \ (n \text{ odd}), \quad e(n) = 2 + 2v_3(n) \ (n \text{ even}).$$

Consequently all labels are odd. Only then use that h_S is a 3-adic unit and $v_3(w) = 1 + v_3(a-1)$ to infer the displayed 3-divisibility.

At 2, with $v(2) = 1$ and $v(\rho) = 1/2$, the retained integral formal series gives $v(z([j]P)) = 3/2 + v_2(j)$ for nonzero integers j. The already established oddness implies $v(h_S) = -3/2$ and $v(w) = v_2(a-1) \geq 1$, so both rational coefficients of w are 2-integral. Also B_0 is 2-integral. The only possibly nonintegral term left in Θ/S_0 is $-6X^2$; its negative valuation cannot cancel the integral terms. This gives $X \in \mathbb{Z}_{(2)}$. The same valuations and the quaternion criterion prove the converse. Zero norm summands and all chart sign choices are covered, and no new square-root branch is introduced.

The supplying arguments are the packed-anchor leading-residue proof, `proofs/prior/round6/arithmetic_round6.tex`, and the actual upstream formal/quaternion passages at lines 8132–8244 and 10973–11024. The conic witnesses are existentially equivalent to (3); no bijection between all witness tuples is claimed.

3.5 Apply the new height bound to the actual consumers

The current input and three scalar contacts are

$$\beta = 2401 + 2754(2X+1)^2 = 11016X^2 + 11016X + 5155, \qquad r = \beta(\beta-1)(32\beta+343).$$

The anchor excludes $X = -1/2$, so β lies in the domain of (2). The current scalar proof supplies $M_{\mathcal{S}}(\beta)^{28} \mid a-1$. With $c = |a-1|$,

$$h(\beta) \leq \mathcal{H} c^{577/896}.$$

The detector poles retain orders $1, 12, 45$, their current contact-unit normalization, and the same three positive square scales $1, C_0^2, (C_0+C_1)^2$. Their unchanged fractional-linear map $\phi(t) = -49 + 1/t$ is defined and avoids the four finite target contacts on the proved source domain. The long logarithm G_{12} still supports order 45 and G_2 supports order 12 at all consumed primes. These are the actual local arguments of the preceding proof and do not depend on the global exponent being nineteen.

For completeness, if the three contact radicals divide a nonzero integer N to order k, their individual height estimates are now bounded by $T_0 = (\mathcal{H} + \log 50)|N|^{\alpha/k}$. The retained scale argument gives $h(C) \leq 3T_0 + (5/2)\log 2 < 7\mathcal{H}|N|^{\alpha/k}$, because $3\log 50 + (5/2)\log 2 < 16$ and $\mathcal{H} \geq 4$. All four labels are already odd, so the inherited product-height identity and root lower bound give

$$B^2 - 6 < 7\mathcal{H}|N|^{\alpha/k}, \qquad B = \max(|n|, |a|, |t|, |b|). \tag{4}$$

This is the product in K, not its rational norm. The optional projective-height improvement is not needed for this selection.

3.6 Exact numerical margins, with a weaker sufficient scalar target

Put

$$c_* = 2 \cdot 3^{1181}, \qquad b_* = c_* - 1.$$

Every solution of the selected anchor has $c \geq c_*$ and $|a| \geq b_*$. We use $h(\beta) < c/4$, in place of the former stronger $c/80$ target, and check the actual final input contradiction below. The exact inequalities

$$3^{53} > 2^{84}, \qquad \mathcal{H} < 3 \cdot 2^{663}$$

give a short transparent certificate. If $L = \log_2 3$, then $\log_2 c \geq 1 + 1181L$ and $\log_2(4\mathcal{H}) < 665 + L$. The difference is positive because

$$319 + (319 \cdot 1181 - 896)\frac{84}{53} - 665 \cdot 896 = \frac{8199}{53} > 0.$$

Hence

$$c^{319/896} > 4\mathcal{H}, \qquad h(\beta) < c/4 \leq (|a| + 1)/4. \tag{5}$$

The independent checker also verifies directly $c_*^{319} > (4\mathcal{H})^{896}$ using integers only.

For the literal Lucas polynomial

$$F = t^2 - a(a-2)tb + (a-2)^2b^2 - 1,$$

the retained bound is $|F| < 5B^4$ for $B > 4$. If $F \neq 0$, apply (4) at order 45. Since $\alpha/45 < 1/2$,

$$B^{143/360} < 7\mathcal{H}5^{\alpha/45} + 6 < 24\mathcal{H}.$$

But $B \geq |a| \geq c/2 \geq 3^{1181}$ and $24\mathcal{H} < 9 \cdot 2^{666}$ give the opposite strict inequality:

$$(143 \cdot 1181 - 720)\frac{84}{53} - 666 \cdot 360 = \frac{1418412}{53} > 0.$$

Equivalently the checker verifies $b_*^{143} > (24\mathcal{H})^{360}$. Thus $F = 0$. The nonidentity root for b gives $b \neq 0$, so the inherited Lucas theorem now gives $\log B > |a| - 3$.

If $n \neq \beta$, its nonzero reduced numerator N satisfies $|N| \leq 2e^{h(\beta)}B$, and the local input comparison gives order 12. Applying (4) and using $\alpha/12 < 2$ yields

$$B^{2-\alpha/12} < 40\mathcal{H}e^{(\alpha/12)h(\beta)},$$

$$\log B < \frac{12}{24-\alpha}\log(40\mathcal{H}) + \frac{\alpha}{24-\alpha}h(\beta).$$

The first inequality follows, for example, from $7\mathcal{H}2^{\alpha/12} + 6 < 40\mathcal{H}$; the added constant is absorbed since $h(\beta) \geq 0$ and $B > 1$. The same exact height bound gives $40\mathcal{H} < 2^{670}$. Since $100 \cdot 670 < b_*$, we have $\log(40\mathcal{H}) < |a|/100$. Finally $\alpha < 19$ gives

$$\log B < \frac{12}{5}\frac{|a|}{100} + \frac{19}{5}\frac{|a|+1}{4} = \frac{487}{500}|a| + \frac{19}{20} < |a| - 3,$$

where the last inequality holds for $|a| > 152$. The anchor supplies that bound. This contradicts the Lucas conclusion. Every use of the old $c/80$ margin in the final soundness proof has therefore been replaced by a checked use of (5); the old numerical inequality is not asserted for this smaller anchor.

Consequently $n = \beta \in \mathbb{Z}$. Together with the anchor's 2- and 3-integrality, this forces $X \in \mathbb{Z}$: $11016 = 2^3 3^4 17$, and a negative valuation of X at any other prime makes the quadratic term of β uniquely minimal, including the leading-coefficient valuation one at 17.

3.7 Completeness, implementation, and verification boundary

The new height estimate is used only in soundness. It changes no point, scale, pole, logarithm, detector contact, or congruence in completeness. The even anchor requires only the finite preparation $3^{1181} \mid a-1$, while the inherited construction allows arbitrary divisibility of the positive scalar index $a-1$. Retain its order of choices: scalar preparation and roots, fixed positive square scales, continuity conditions, then separated contact primes and the primary progression $\Pi^{44} \mid h$. The scalar pole preparation, Gaussian positivity witnesses, both-place formal arguments and local avoidance conditions remain those of the selected local proofs.

No coordinate is added by changing the fixed integer S_0. Its even packing coefficient remains three, so the symbolic physical anchor equations are unchanged. The convenient identity $3^{2364} = (3^{394})^6$ supports a separately audited input-fixed diagonal coefficient change. It is that separate algebraic certificate, not this height argument, which proves any reported coefficient maximum.

The finite checker verifies the source anchor and input equations, local seed calculations, Euler-product certificate, explicit constants, and all three selected numerical consumers. The complete scalar/detector proofs and the inherited global arithmetic still require their stated mathematical review. This is a conditional reapplication with a smaller height exponent and an explicit valid parameter choice; no optimality, whole-proof formalization, or historical priority claim is made.

4 Two-prime coefficient balancing without new variables

The selected source has 389 rational coordinates, including the input X, and 357 quadratic residuals. Its fixed anchor is

$$S_0 = 3^{2364} = c^6, \qquad c = 3^{394}.$$

The preceding arithmetic argument supplies this parameter. The present change is an exact rational bijection on the already constructed system; its zero-set equivalence needs no further arithmetic input.

4.1 The explicit diagonal map

Write x_i for an old compiled coordinate and y_i for its replacement. Set

$$x_i = 2^{b_i} 3^{a_i} y_i.$$

The input has $a_0 = b_0 = 0$. All unlisted coordinates also have both exponents zero. The complete nontrivial map is as follows:

Stored source coordinates	Number	$a_i/394$	b_i
The two coefficients of h_S	2	1	0
The stored real coefficient of $x(S)$	1	1	1
The stored ρ coefficient of $x(S)$	1	1	0
The two coefficients of $f(S-P)$	2	1	0
The two coefficients of $w = h_S\tau$	2	2	0
Translated anchor witnesses U, V, W	3	4	0
Materialized physical anchor Θ	1	4	0
Stored coordinate for the first nontrivial height scale	1	0	-21

The last row denotes precisely the coordinate named `heightProduct.scale0.quadratic`. The additional dyadic factor in the second row targets precisely `S.chart.x.0.quadratic`. Names and source identifiers are bound in the certificate. Semantic height scales and point functions are recovered through the original value map after decoding these stored coordinates. In this source the semantic first height scale is $A = C_0^2 = 4x_s$, where x_s is the stored coordinate in the last row. Thus after the change $A = 2^{-19} y_s$, and $y_s = 2^{19} C_0^2$ need not itself be a rational square. Every detector square-scale and parity argument applies to the decoded $A = C_0^2$. The transformed equations retain precisely that value; no square property is imposed on the new coordinate y_s.

There are thirteen changed coordinates: twelve with a ternary factor and two with a dyadic factor, with one overlap. Every factor is a fixed positive nonzero rational number, so the displayed map is a bijection on $\mathbb{Q}^{389}$ and fixes X. Its inverse divides each coordinate by the same factor. It introduces no new quantified variable or choice of a root.

The transformed chart equations carry their corresponding powers of two and three. In particular the coordinates for h_S and $x(S)$ have different semantic degrees but are scaled as displayed; decoding the whole transformed system recovers the original identity $x(S) = h_S^2$.

4.2 Rational row normalization and the zero projection

For an integer quadratic source residual $R_j(x, S_0)$, let

$$\widetilde{R}_j(y) = R_j((2^{b_i} 3^{a_i} y_i)_i, 3^{2364}).$$

Its coefficients are rational because some b_i are negative. Let $L_j > 0$ be the least common multiple of their reduced denominators, and let $g_j \neq 0$ be the signed integer content of $L_j \widetilde{R}_j$, with sign chosen to make the first coefficient in canonical monomial order positive. Put

$$d_j = g_j / L_j \in \mathbb{Q}^*, \qquad Q_j = \widetilde{R}_j / d_j \in \mathbb{Z}[y_0, \ldots, y_{388}].$$

Each Q_j is primitive and has degree at most two. The selected quartic is

$$P_{389}(y) = \sum_{j=1}^{357} Q_j(y)^2.$$

Every d_j is nonzero. Consequently, over $\mathbb{Q}$,

$$P_{389}(y) = 0 \iff Q_j(y) = 0 \text{ for all } j \iff R_j((2^{b_i} 3^{a_i} y_i)_i, 3^{2364}) = 0 \text{ for all } j.$$

Together with the diagonal bijection, this proves equality of the existential projections to X. Rowwise rescaling preserves the common zero set; it need not make the new sum of squares one scalar multiple of the preceding sum of squares.

The rational divisors matter. In the selected system five d_j are nonintegers, with reduced denominators 2^{21} three times, 2^{42} once, and 2^{13} once. The certificate records every divisor's numerator and positive denominator. The checker reconstructs each substituted row and verifies the exact rational proportionality and primitive integer normalization, including these five cases.

4.3 Where the coefficient saving comes from

The ternary part is the preceding coefficient balancing: the affected rows have a span at most two in their powers of c, so their squares involve at most c^4. With only that map, the large coefficient of the stored height-scale coordinate in the physical anchor is $201326592c^2 = 3 \cdot 2^{26} c^2$. The extra factor 2^{-21} changes it to $96c^2$. Its squared contribution is then $9216c^4$.

The extra factor two on the stored real coefficient of $x(S)$ balances the two shifted-point equations after primitive row normalization. The complete exact expansion, including contributions shared by different rows, gives

$$\boxed{\max |\operatorname{coeff}(P_{389})| = 28902c^4 = 28902 \cdot 3^{1576}.} \tag{6}$$

This integer has 757 decimal digits and 2513 bits. The quartic has 26,295 nonzero monomials, degree exactly four, content one, and coefficient one on X^4. All 389 coordinates occur. Every coefficient in the delivered text table is written as an ordinary decimal integer.

For comparison on the preceding anchor alone, the same two dyadic factors lower its exact maximum from $1 + 40532396646334464 \cdot 3^{1860}$ to $28902 \cdot 3^{1860}$, reducing 905 digits to 892 without changing its coordinate count. The further decrease to 757 digits in (6) comes from the independently justified anchor exponent 2364.

A linear optimization suggested the factors; its floating-point output is not a verification dependency or an optimality certificate. The delivered producer uses the displayed integer exponents and exact rational arithmetic. A separately authored checker imports no producer or compiler: it recovers row normalization by pivot ratios and common denominators, verifies every nonzero divisor, expands the sum of squares by unordered term pairs, and compares every coefficient in both the JSON polynomial and the decimal text table. The source gates and earlier compilation are covered by their separate exact checks.

This establishes the finite transformation and coefficient assertions. The interpretation of the original source's projection as the integers retains the global arithmetic qualifications listed in the dependency inventory.

5 Exact verification and a portable review workflow

The source compiler reports 339 gates. The separate gate checker verifies all 339 gate identities and 330 compilation operations. The detector checker reconstructs all sixteen semantic tests and their orders, the actual products, norm and pole consumers, and passes 384 source assertions, seven Laurent identities, three normalization identities and two uniform-pair identities. A separately authored conservation checker verifies 157 assertions, including semantic agreement of 362 unchanged equations with the supplied byte-identical 392-variable reference builder. There is no further saving from the final elimination search.

The arithmetic source checker verifies 97 exact source and numerical conditions. A separate arithmetic certificate checks 67 inequalities and normalizations, without importing the author checker. The written review also checks the actual pinned height proof, the genus and level conventions, small-prime subset/tail argument, noncircular anchor argument and all three height consumers. The numerical certificates do not by themselves prove these uniform or inherited global assertions.

The separately authored coefficient checker reconstructs all 357 primitive rows, every rational row divisor, the rational inverse coordinate map, the full quartic sum of squares and all 26,295 decimal coefficients. Six negative probes, including altered coefficients with rebound advertised digests, were rejected. The complete table SHA256 is

`1c612d20915667dfd81a7895f8d93231d5efeaf86ecfaa15194408d298d35573`

The other source, scalar, gate, elimination, stage-bridge, parameter-binding and independent full-table audits remain separate checks.

The rebuild uses only Python 3.11+ and Node.js standard libraries and needs no network. The wrapper starts from a fresh directory seeded with exactly the declared source and proof inputs. It regenerates all mathematical artifacts, compares deterministic file hashes and the committed decimal table, checks all required successful audit reports, and verifies that both the original sources and every copied build input remain unchanged.

Completion is a structured atomic record, bound to the executed pipeline hash and its exact ordered list of 23 successful stages. The final stdout line is diagnostic. This change addresses a weakness exposed when the preceding wrapper once failed to observe that line despite matching mathematical outputs. The cause of that historical stdout event remains unknown; it is preserved as evidence rather than given a speculative explanation. Forty-five focused orchestration probes and seven separate adversarial probes check the wrapper's acceptance and failure behavior. These mocks supplement an actual fresh full rebuild; they do not replace it.

The draft package supplies a full patch against the pinned upstream and an incremental patch against the exact preceding 392-variable overlay. Apply one appropriate route and run `reproduce.py--check-source`, then `reproduce.py`. The handoff also runs the repository audit-history checker and preserves its result. A failed equation, changed domain, source mismatch or coefficient mismatch must be investigated rather than resolved by changing golden hashes to whatever happened to be produced.

No public branch, PR, message, merge or release was created by this work. Official manuscript integration would require explicit replacement arguments at the consuming passages and a fresh review under the repository's protocol. The package leaves that decision and any publication to the human submitter and maintainers.